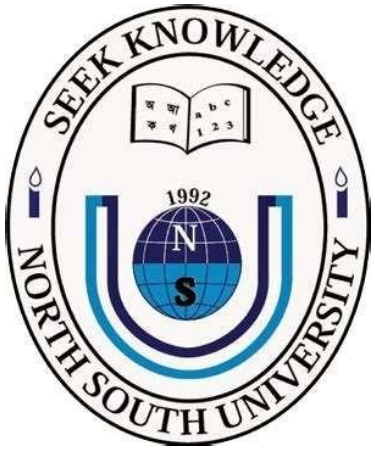
Analog Electronics II: Op-amp & 555 Timer



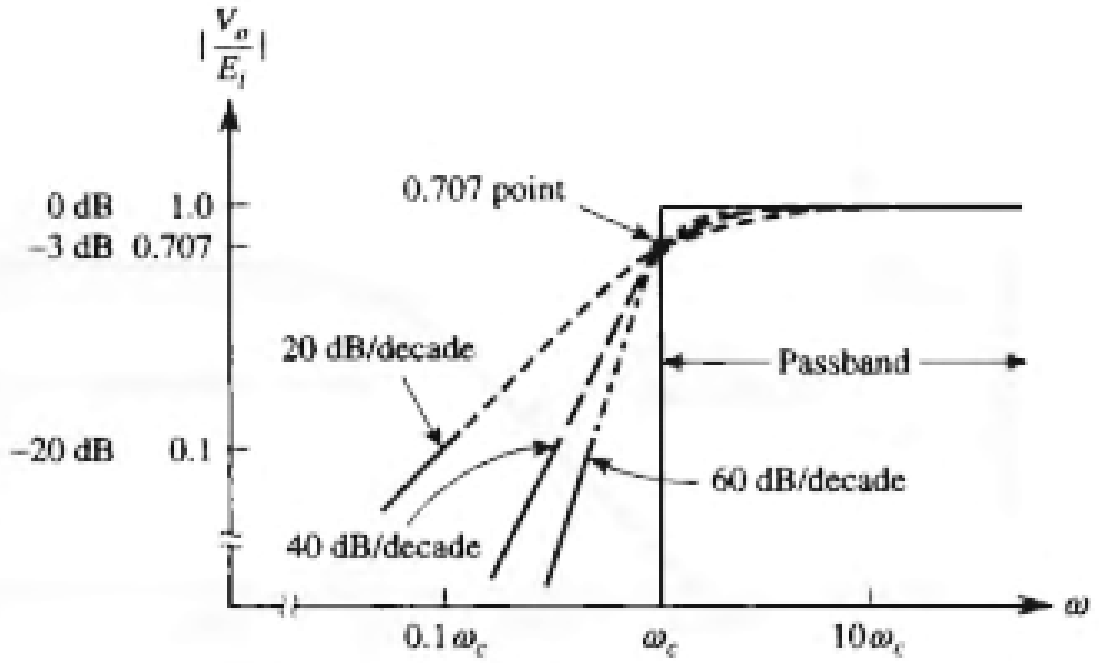
Course instructor:
Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

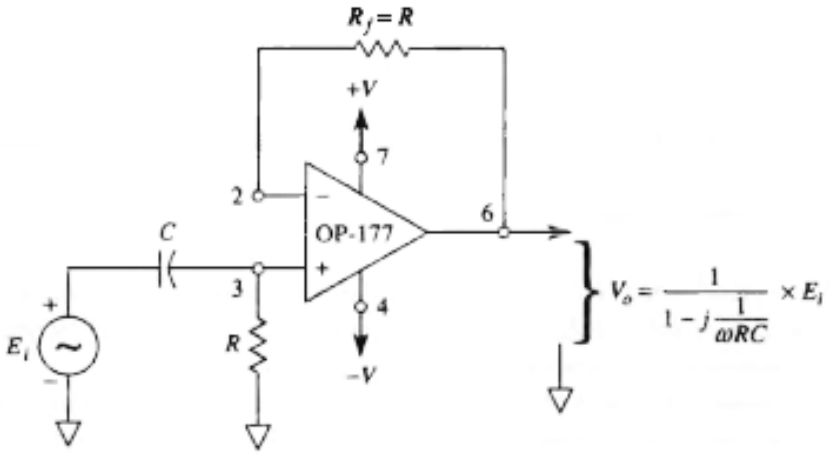
Lecture # 11

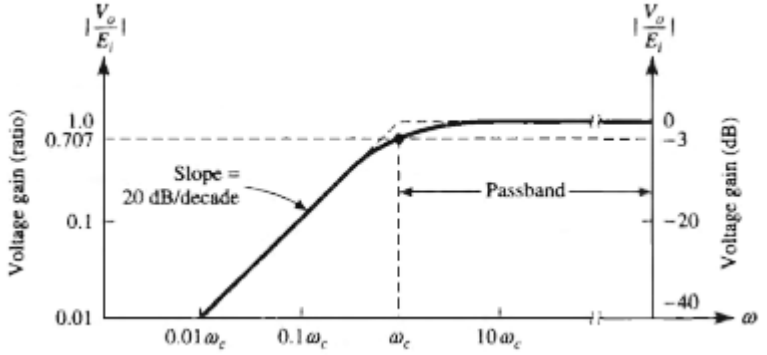
Active Filters: First-Order Highpass Filter

HIGH-PASS BUTTERWORTH FILTERS



20-dB/Decade Filter

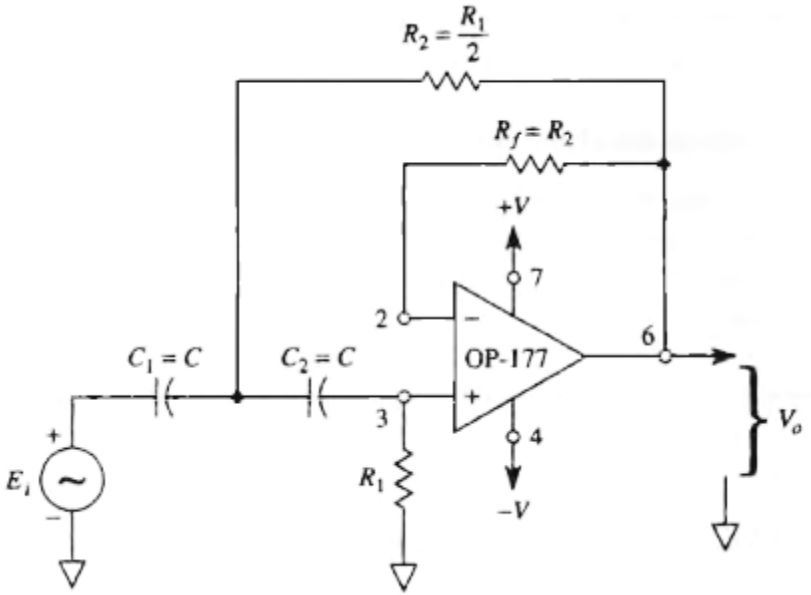


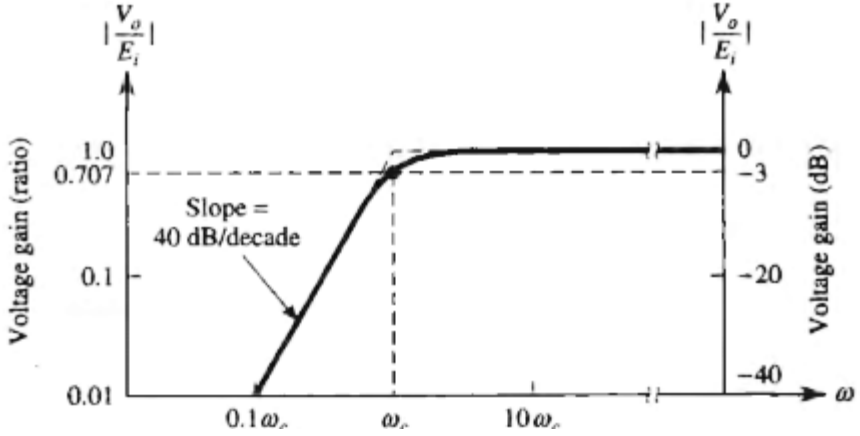


Design procedure for 20-dB/decade high-pass

1. Choose the cutoff frequency, ω_c or f_c.
2. Choose a convenient value of C, usually between 0.001 and 0.1 μF.
3. Calculate R from $\omega_c = \dfrac{1}{RC} = 2\pi f_c$
4. Choose $R_f = R$.

40-dB/Decade Filter





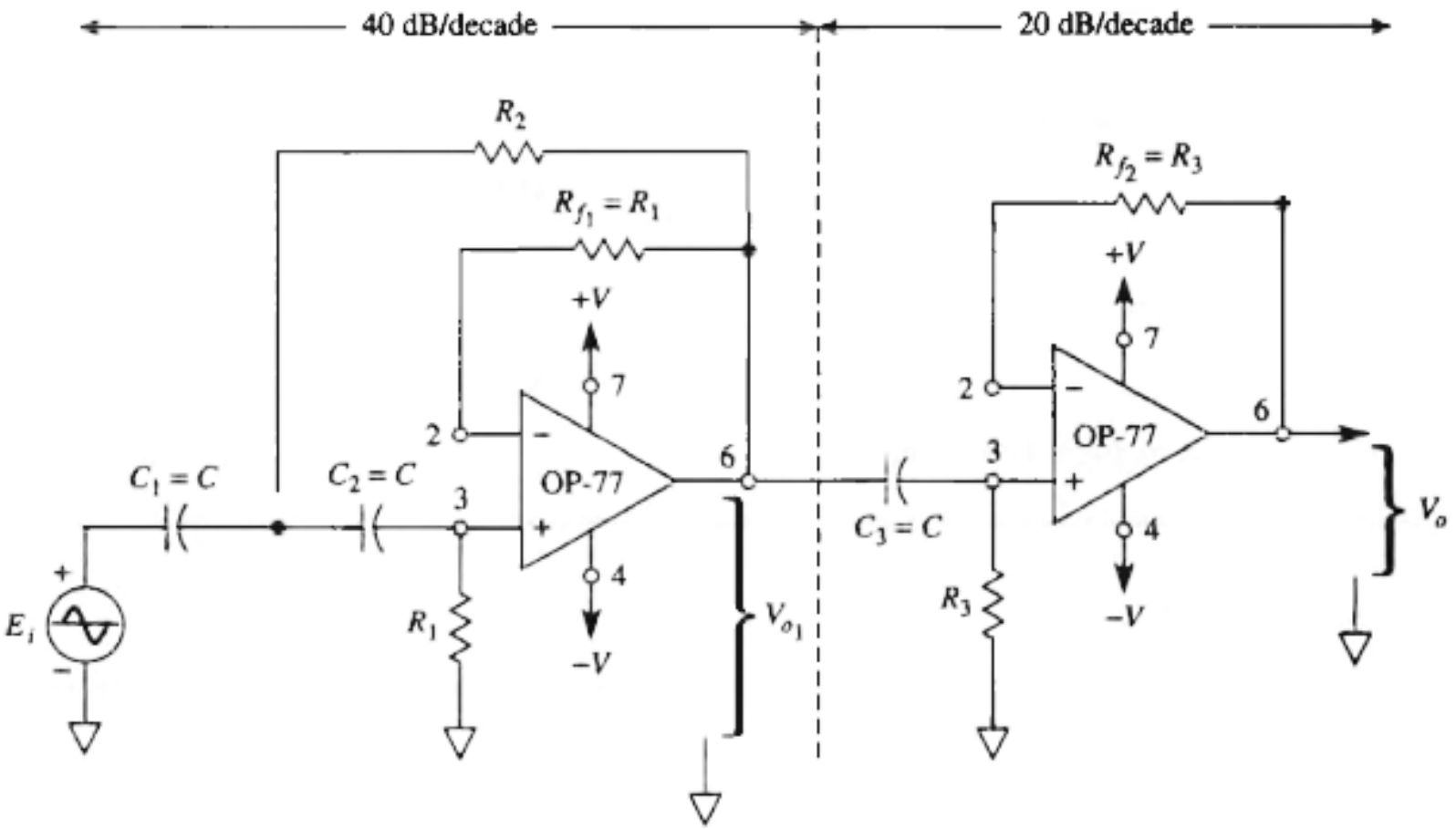
Design procedure for 40-dB/decade high-pass

1. Choose a cutoff frequency, ω_c or f_c.
2. Let $C_1 = C_2 = C$ and choose a convenient value.
3. Calculate R_1 from

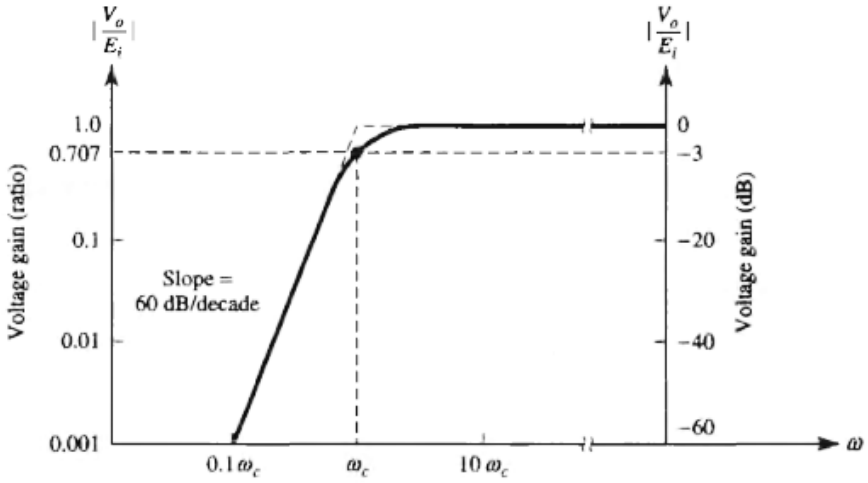
$$R_1 = \frac{1.414}{\omega_c C}$$

4. Select $R_2 = \frac{1}{2}R_1$
5. To minimize dc offset, let $R_f = R_1$.

60-dB/Decade Filter



60-dB/Decade Filter



Design procedure for 60-dB/decade high-pass

1. Choose the cutoff frequency, ω_c or f_c.
2. Let $C_1 = C_2 = C_3 = C$ and choose a convenient value between 100 pF and 0.1 μF.
3. Calculate R_3 from

$$R_3 = \frac{1}{\omega_c C}$$

4. Select

$$R_1 = 2R_3$$

5. Select

$$R_2 = \tfrac{1}{2}R_3$$

6. To minimize dc offset current, let $R_{f_1} = R_1$ and $R_{f_2} = R_3$.

COMPARISON OF $|A_{CL}|$

ω	20 dB/decade; Fig. 11-7(a)	40 dB/decade; Fig. 11-8(a)	60 dB/decade; Fig. 11-9(a)
$0.1\omega_c$	0.1	0.01	0.001
$0.25\omega_c$	0.25	0.053	0.022
$0.5\omega_c$	0.445	0.24	0.124
ω_c	0.707	0.707	0.707
$2\omega_c$	0.89	0.97	0.992
$4\omega_c$	0.97	0.998	0.999
$10\omega_c$	1.0	1.0	1.0

Example 11-10

For Fig. 11-9(a), let $C_1 = C_2 = C_3 = C = 0.1\ \mu$F. Determine (a) R_3, (b) R_1, and (c) R_2 for $\omega_c = 1$ krad/s. ($f_c = 159$ Hz.)

Solution (a) By Eq. (11-11),

$$R_3 = \frac{1}{(1 \times 10^3)(0.1 \times 10^{-6})} = 10\ \text{k}\Omega$$

(b) $R_1 = 2R_3 = 2(10\ \text{k}\Omega) = 20\ \text{k}\Omega$.
(c) $R_2 = \frac{1}{2}R_3 = \frac{1}{2}(10\ \text{k}\Omega) = 5\ \text{k}\Omega$.

Example 11-11

Determine (a) R_3, (b) R_1, and (c) R_2 in Fig. 11-9(a) for a cutoff frequency of 60 kHz. Let $C_1 = C_2 = C_3 = C = 220$ pF.

Solution (a) From Eq. (11-11),

$$R_3 = \frac{1}{(6.28)(60 \times 10^3)(220 \times 10^{-12})} = 12\ \text{k}\Omega$$

(b) $R_1 = 2R_3 = 2(12\ \text{k}\Omega) = 24\ \text{k}\Omega$.
(c) $R_2 = \frac{1}{2}R_3 = \frac{1}{2}(12\ \text{k}\Omega) = 6\ \text{k}\Omega$.

If desired, the 20-dB/decade section can come before the 40-dB/decade section, because the op amps provide isolation and do not load one another.

Practice Problem

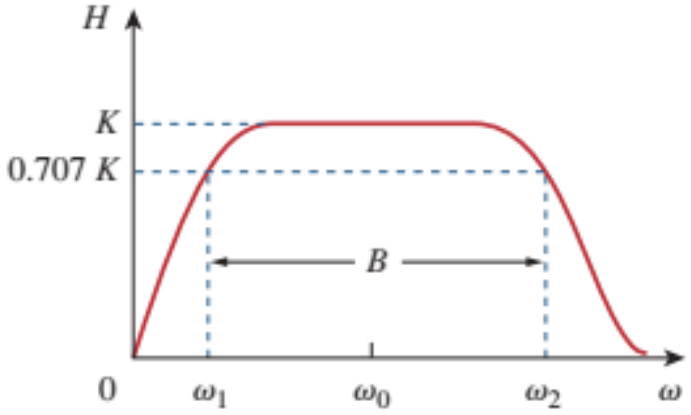
An Electrocardiogram (ECG or EKG) is a medical test that records the electrical activity of the heart over time. It helps diagnose heart conditions by analyzing the heart's rhythm, electrical conduction, and abnormalities in heart function.
ECG electrodes placed on the skin detect electrical activity from the heart. The raw signal contains useful heart rhythms (0.5 Hz – 100 Hz) but also high-frequency noise (>100 Hz) from muscle activity (EMG) and electrical interference (60 Hz power lines).
Design a Butterworth high pass filter with 40 dB/decade slope on the frequency response. The minimum value of the resistances should be 1 kΩ.

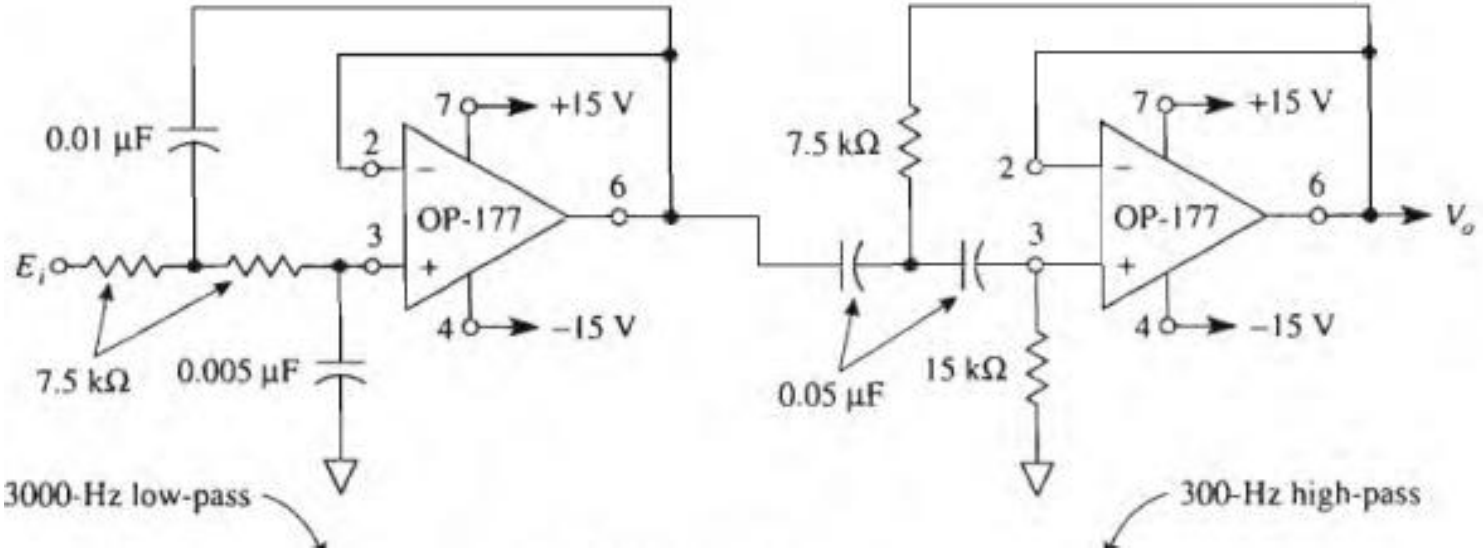
Solution:

Bandpass Filter

Wideband Filter Circuit







Bandpass Filter

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \left(-\frac{1}{1 + j\omega C_1 R}\right)\left(-\frac{j\omega C_2 R}{1 + j\omega C_2 R}\right)\left(-\frac{R_f}{R_i}\right)$$

$$= -\frac{R_f}{R_i}\frac{1}{1 + j\omega C_1 R}\frac{j\omega C_2 R}{1 + j\omega C_2 R}$$

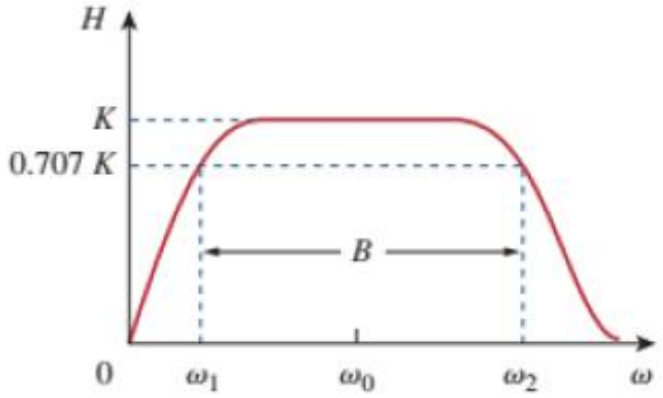
The lowpass section sets the upper corner frequency $\omega_2 = \dfrac{1}{RC_1}$

the highpass section sets the lower corner frequency $\omega_1 = \dfrac{1}{RC_2}$

the center frequency $\omega_0 = \sqrt{\omega_1 \omega_2}$

bandwidth $B = \omega_2 - \omega_1$

quality factor $Q = \dfrac{\omega_0}{B}$



Bandpass Filter

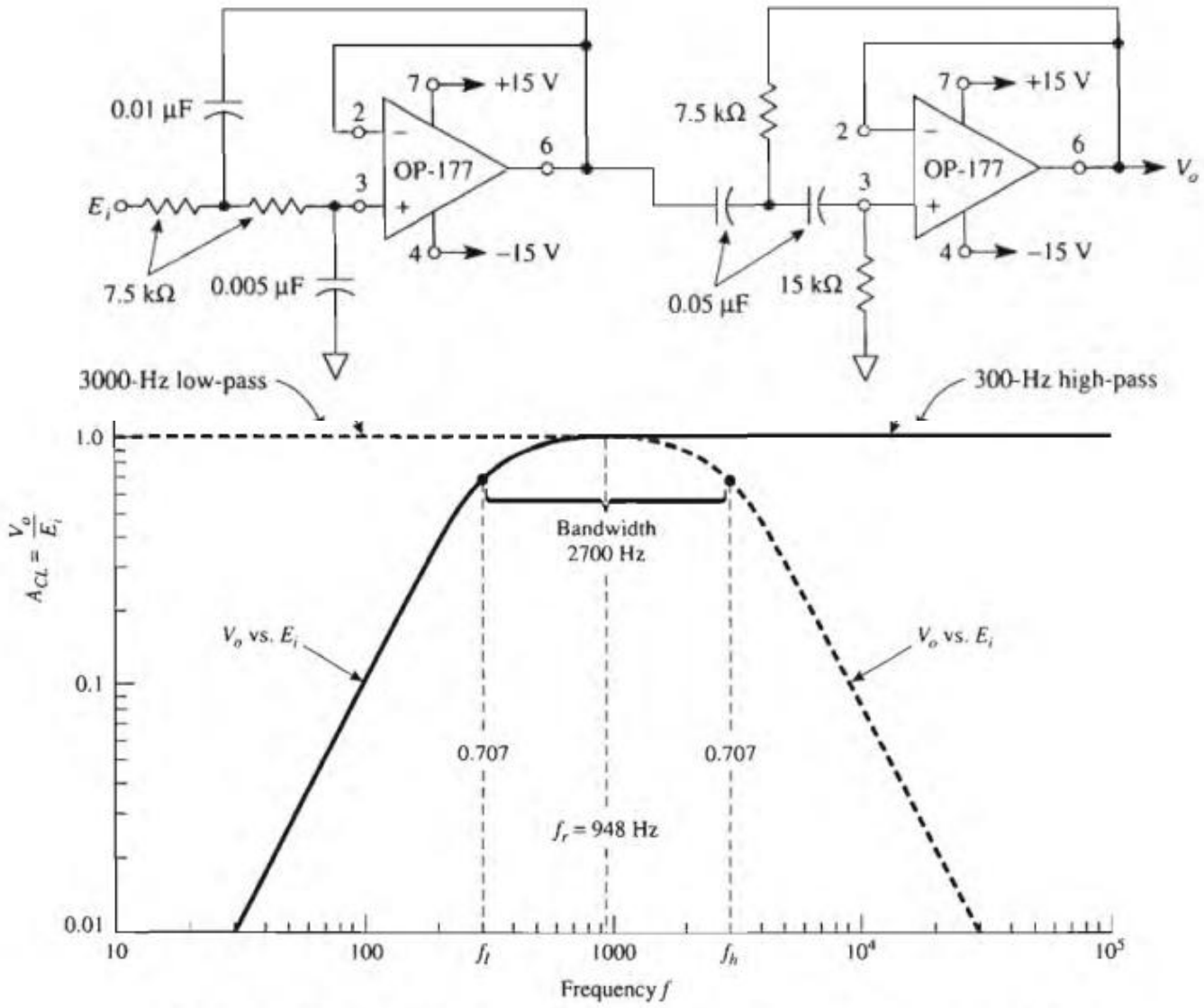
$$\mathbf{H}(\omega) = -\frac{R_f}{R_i}\frac{j\omega/\omega_1}{(1 + j\omega/\omega_1)(1 + j\omega/\omega_2)} = -\frac{R_f}{R_i}\frac{j\omega\omega_2}{(\omega_1 + j\omega)(\omega_2 + j\omega)}$$

At the center frequency $\omega_0 = \sqrt{\omega_1\omega_2}$, the magnitude of the transfer function is

$$|\mathbf{H}(\omega_0)| = \left|\frac{R_f}{R_i}\frac{j\omega_0\omega_2}{(\omega_1 + j\omega_0)(\omega_2 + j\omega_0)}\right| = \frac{R_f}{R_i}\frac{\omega_2}{\omega_1 + \omega_2}$$

Thus, the passband gain is $K = \dfrac{R_f}{R_i}\dfrac{\omega_2}{\omega_1 + \omega_2}$

Wideband Filter



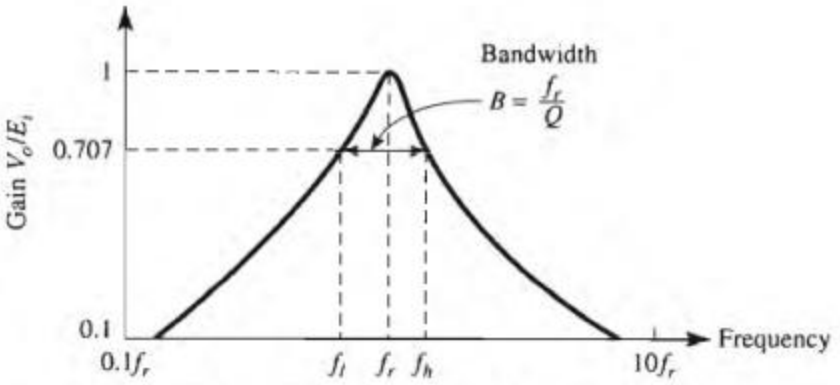
Practice Problem

For a music production, a particular frequency range 250 Hz- 3000 Hz needs to be filtered in order to eliminate noise and other interfering sounds. Design an active bandpass filter.
Design Requirements:

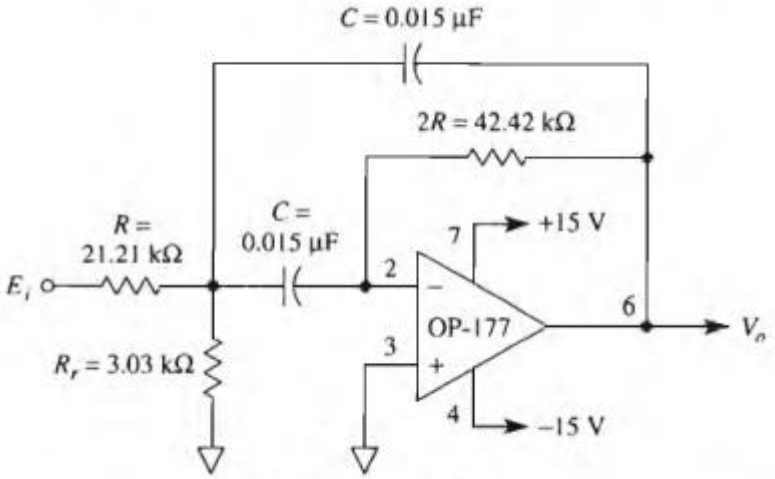
Passband frequencies: between 250 Hz and 3,000 Hz
Passband Gain = 10
Input impedance R= 10 kΩ.
Draw the circuit diagram and find out the required values of corresponding circuit components.

Solution:

NARROWBAND BANDPASS FILTERS



(a) Typical frequency response curve of a narrowband filter.



(b) Narrowband filter.

FIGURE 11-12 Narrow bandpass filter circuit and its frequency response for the component values shown; $f_r = 100$ Hz, $B = 500$ Hz, $Q = 2$, $f_l = 780$ Hz, and $f_h = 1280$ Hz. (a) Typical frequency response of a bandpass filter; (b) narrow bandpass filter circuit.

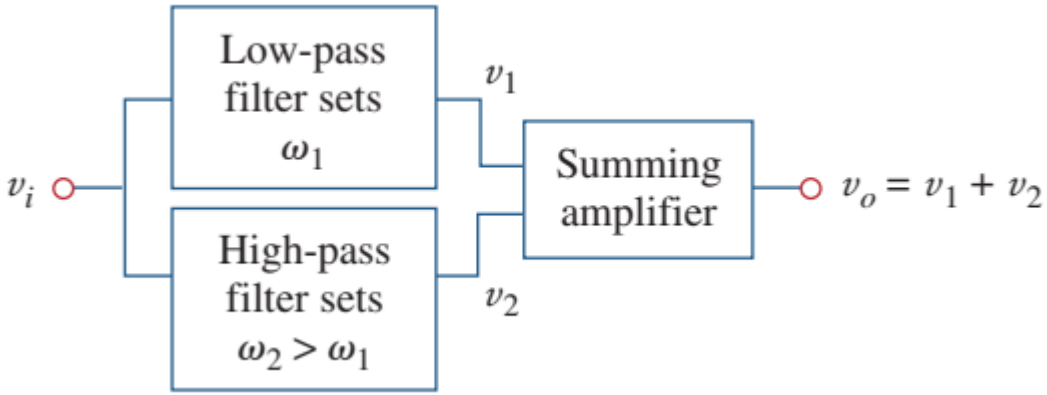
Design procedure

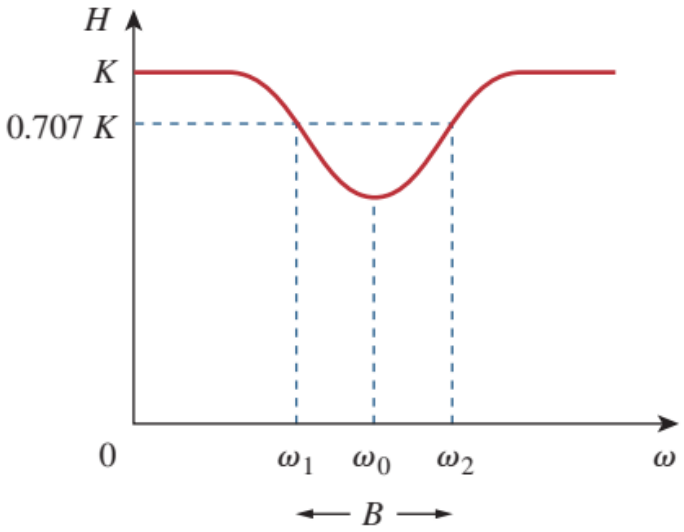
$$B = \frac{0.1591}{RC} \qquad R_r = \frac{R}{2Q^2 - 1}$$

$$B = \frac{f_r}{Q} \qquad f_r = \frac{0.1125}{RC}\sqrt{1 + \frac{R}{R_r}}$$

Active Filters: Bandreject (or Notch) Filter

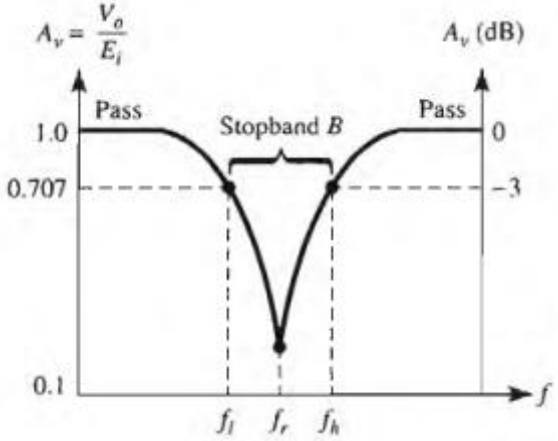
Wideband Filter Circuit





NOTCH FILTERS

NARROWBAND BANDREJECT FILTER



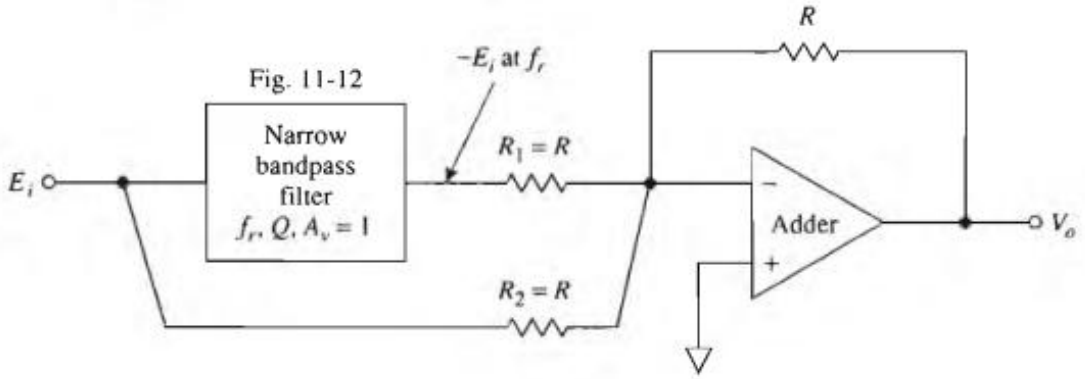


FIGURE 11-14 A notch filter is made by a circuit that subtracts the output of a bandpass filter from the original signal.

Procedure to Make a Notch Filter

1. Make a bandpass filter that has the same resonant frequency, bandwidth, and consequently Q as the notch filter.
2. Connect the inverting adder of Fig. 11-15 by selecting equal resistors for R. Usually, $R = 10\ \text{k}\Omega$. (A practical fine-tuning procedure is presented in the next section.)

NOTCH FILTERS

NARROWBAND BANDREJECT FILTER

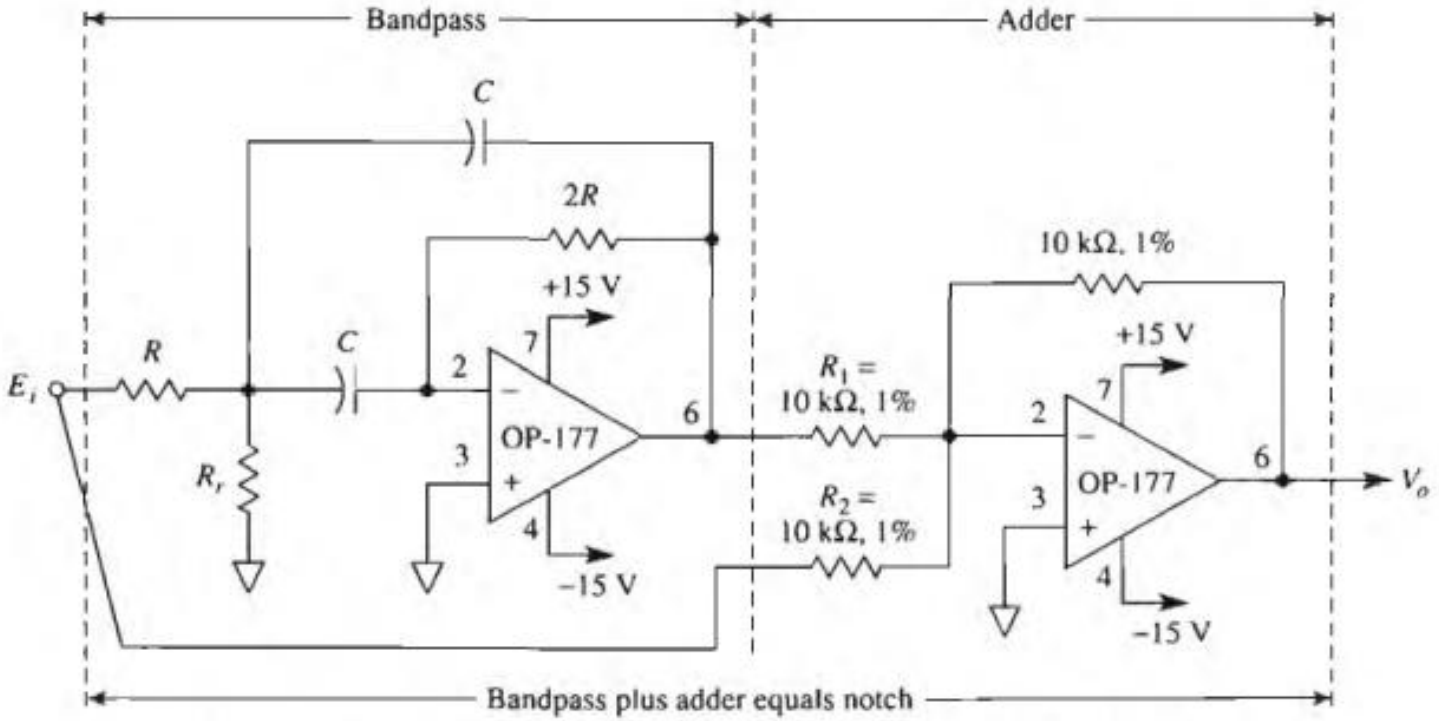


FIGURE 11-15 This two-op-amp notch filter is made from a bandpass filter plus an inverting adder. If $C = 0.33\ \mu F$, $R = 40.2\ k\Omega$, and $R_f = 201\ \Omega$, the notch frequency will be 120 Hz and reject a bandwidth of 12 Hz.

Practice Problem

Design a **biomedical signal conditioning circuit** for an **EEG amplifier** that picks up unwanted **50 Hz power-line interference**.
To remove this narrowband interference without affecting nearby signal frequencies, you decide to implement a **narrowband band-reject filter** using a **band-pass–plus–subtractor method**.

Design Specifications:

- Interference (resonant) frequency, f_0=50 Hz
- Desired quality factor, Q=25
- Passband gain of the final output = 1 (unity)
- Supply: ±15 V op-amp system
- Available passive components: capacitors (100 pF – 0.1 µF)

Solution:

References:

1. Butterworth Filters:
Low-Pass, High-Pass, Wideband/Narrowband Bandpass, Wideband/Narrowband Bandreject Filters.

-Operational Amplifiers and Linear Integrated Circuits- by Robert F. Coughlin & Frederick F. Driscoll, **Chapter 11**